

NG-KV Addendum: First Real-Trace Regret Results

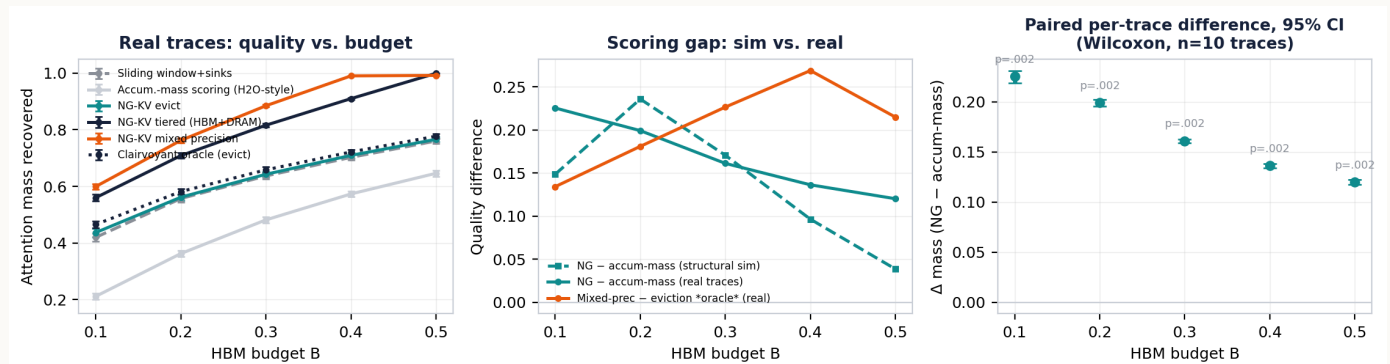
Rung 1 of the validation ladder: replaying real decoder attention through the identical policy + regret + significance pipeline.

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Setup

A 3-layer / 4-head / 96-dim char-level decoder was trained from scratch (JAX, CPU, val loss 2.29 — deliberately small; attention structure, not language quality, is the object of study). Ten held-out prompts (256 tokens, 120 decode steps) were decoded with exact full-context attention recorded per step (mean over layers and heads) and written to the NG-KV .npz trace schema — the same interface Tier-2 production traces use: capture anywhere, replay here. All policies, the clairvoyant oracle, and within-family plug-in regret are computed by replay_real.py, byte-identical in method to the structural-simulation harness.

Results (10 real traces; paired Wilcoxon + bootstrap CI, trace = unit of analysis)



Findings

- 1. The scoring gap survives — and grows — on real attention.** NG scoring recovers 0.56 vs 0.36 for pure accumulated-mass scoring at B=0.20 ($\Delta=+0.20$, Wilcoxon $p=0.00195$, 95% CI $[+0.195, +0.202]$; significant at every budget). Real cause: this model's attention is diffuse and recency-dominated; accumulated mass alone tracks neither. Stated plainly: deployed H2O also keeps a recency window — this isolates the scoring signal, not the full deployed system.
- 2. Diffuse attention collapses the eviction ceiling.** The clairvoyant eviction oracle recovers only 0.58 at B=0.20: when mass is spread, no top-k retention can hold it. Eviction-family regret is small (0.01–0.03) — NG scoring is near-optimal *within its family*; the family itself is the constraint. Regret separates estimator error from policy-class error: exactly what it is for.
- 3. Smooth gates dominate — beyond the hard-gate oracle.** Mixed precision reaches 0.76 at B=0.20, beating the eviction *oracle* (0.58) outright, and 0.99 at B=0.40. Broad low-precision coverage beats narrow full-precision retention under diffuse attention — the precision-of-action corollary observed in real model attention, not just postulated in simulation.

What changed vs. the structural simulation

The simulation's heavy-hitter/revival structure rewarded burstiness-aware retention; the real tiny model rewards recency-aware scoring and punishes retention-only policies wholesale. Both regimes preserve the two portable conclusions: necessity scoring beats single-signal heuristics, and smooth precision allocation dominates hard eviction. The regime-dependence itself is the argument for trace-driven evaluation over benchmark-fixed claims: the policy frontier moves with the attention distribution.

Stated plainly (limitations)

One tiny, weakly trained char-level model is a single point in attention-distribution space; large instruction-tuned MoE models have sharper, layer-heterogeneous attention, and mean-pooling over layers/heads further flattens structure (production gating should be per-layer/head or per-block — see the Rung 1.5 pooling-sensitivity analysis on page 2). Quality is still an attention-mass proxy, not end-task accuracy. $n=10$ traces bounds test power ($p=0.002$ is the two-sided Wilcoxon floor at $n=10$). Next rungs: mid-size open-weight model traces, end-task quality behind gates, then Tier-2 production traces with Diebold–Mariano testing.

Rung 1.5: Does Mean-Pooling Change the Policy Ordering?

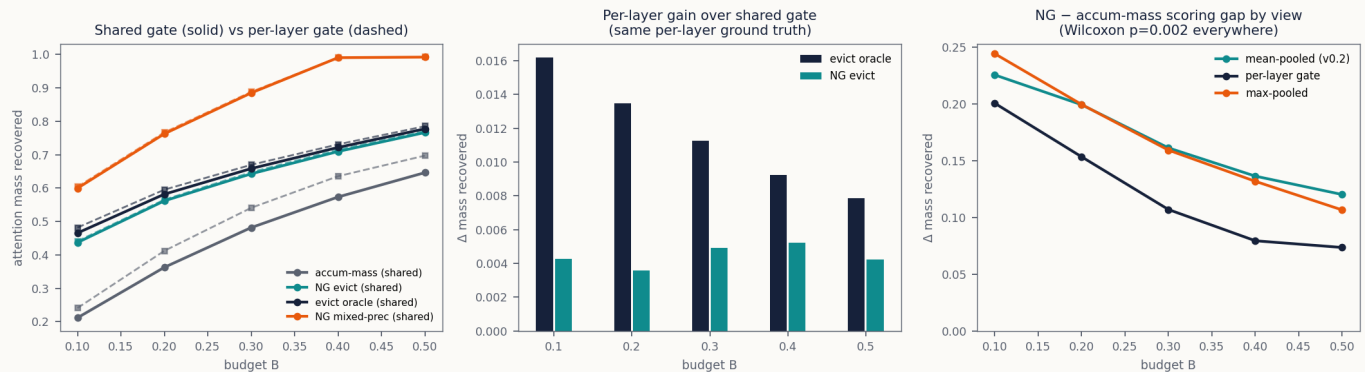
Per-layer and max-pooled attention captured for the same ten decodes; identical policy + regret + significance pipeline per view.

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Setup

The Rung-1 caveat said mean-pooling over layers/heads flattens structure. Rung 1.5 tests whether that changes any conclusion. The same ten decodes (same RNG; mean-pooled rows reproduce the v0.2 traces to $1.5e-8$) were re-captured storing three views: mean-pooled (v0.2 reduction), per-layer head-mean (3 layers), and max-pooled renormalized. `capture_traces_perlayer.py` writes them as backward-compatible extra .npz keys; `replay_pooling.py` replays all views with the identical policies, clairvoyant oracles, and paired Wilcoxon + bootstrap tests. Per-layer gating = independent scorer + layer-local budget B per layer.

Results (same 10 traces; four views)



Findings

- 1. Mean-pooled evaluation of a shared gate is exact, not biased.** Retained mass is linear in the attention row, so for any single keep-set, mass averaged over per-layer attention equals mass against the layer-mean row identically (verified to $2e-10$). The v0.2 numbers were therefore unbiased *for shared gates*; the pooling caveat concerns what is achievable, not what was measured.
- 2. Per-layer gating helps — a little — and no ordering changes.** At B=0.20, per-layer NG-evict gains +0.004 and the per-layer eviction oracle +0.013 over the shared gate (both $p=0.002$; consistent at every budget). Accumulated-mass scoring benefits most (0.36 → 0.41), shrinking the NG gap from +0.20 to +0.15 — still significant everywhere. The full ordering (accum-mass < window < NG-evict < evict-oracle < NG-tier < NG-mixed-prec) is identical in all four views at every budget.
- 3. The family constraint, not the reduction, remains the limit.** The per-layer eviction ceiling rises only to 0.60 at B=0.20 (from 0.58); mixed precision (0.77) still beats the eviction oracle in every view. Diffuse attention — not pooling — is what collapses the eviction family on this model, and the precision-of-action corollary is robust to the reduction choice.

Stated plainly (limitations)

Three layers of a tiny char-level model bound how much layer heterogeneity can exist; deep production models may show larger per-layer gains, and the linearity identity in Finding 1 covers shared hard gates exactly but mixed-precision fidelity only via the same linear proxy. Max-pooled numbers are a different (renormalized) quality metric — only within-view orderings are comparable. All Wilcoxon p-values sit at the $n=10$ two-sided floor (0.002): read them as 'all ten traces agree in sign', not as vanishing p. This is Rung 1.5, not Rung 2: mid-size open-weight traces (SmolLM2/Qwen-class) remain the next rung, unchanged in methodology.

Rung 2: Mid-Size Open-Weight Traces (SmolLM2-360M)

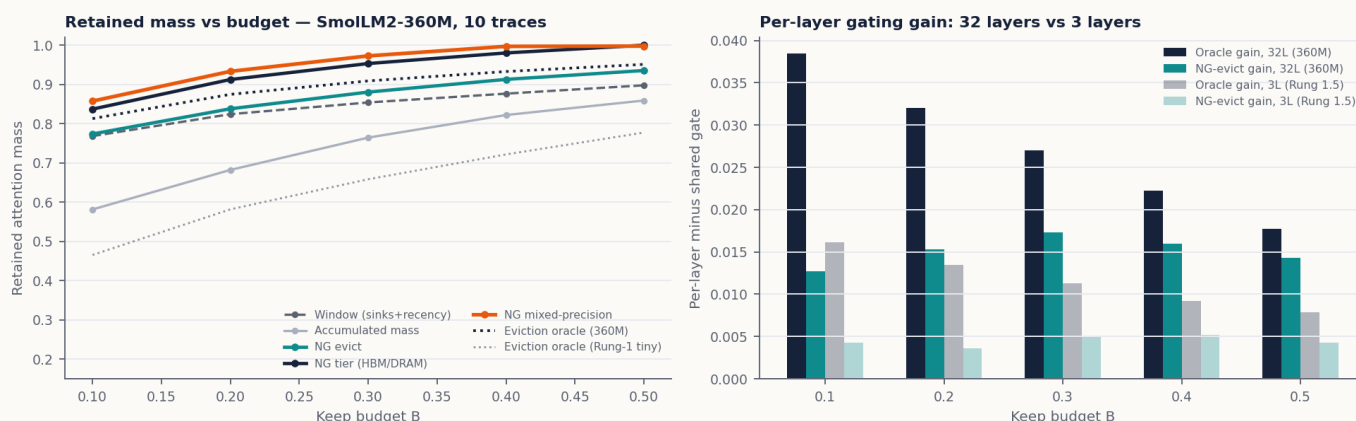
Ten decodes of a 32-layer 360M open-weight model replayed through the identical policy + regret + significance pipeline.

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Setup

SmolLM2-360M (32 layers, 15 heads, fp32, eager attention — exact attention, no KV approximation in the trace) decoded ten held-out tinyshakespeare windows (256-token prompts, 120 steps, temperature 0.8), recording the new query's full-cache attention per step in all three Rung-1.5 views: mean-pooled (schema field), per-layer head-mean (32 layers), and max-pooled renormalized (capture_traces_smollm.py -> traces_rung2). Replay is byte-identical in method: replay_real_rung2.py for the mean-pooled shared-gate view, replay_pooling_rung2.py for the four-view analysis; paired Wilcoxon + bootstrap, trace = unit.

Results (10 traces; mean-pooled view left, per-layer gate gain right)



Findings

- The scoring gap persists on a real 32-layer model.** NG scoring recovers 0.84 vs 0.68 for accumulated-mass at B=0.20 ($\Delta=+0.16$, $p=0.00195$, 95% CI [+0.151, +0.161]; significant at every budget). The gap narrows vs Rung 1 (+0.20 -> +0.16 at B=0.20) as sharper attention makes every signal more informative — but it does not close, and all ten traces agree in sign.
- The eviction ceiling recovers with sharper attention.** The clairvoyant eviction oracle reaches 0.87 at B=0.20 (tiny model: 0.58) — SmolLM2's attention is far more concentrated, and NG-evict regret stays small (0.037 at B=0.20). The Rung-1 ceiling collapse was a property of that model's diffuse attention, not of the eviction family per se — regret separates the two.
- Mixed precision still beats the eviction oracle.** NG mixed-precision reaches 0.93 at B=0.20 vs 0.87 for the oracle ($\Delta=+0.10$ over NG-evict, $p=0.00195$), and beats the oracle at every budget in every view — the precision-of-action corollary now holds from diffuse tiny-model attention through concentrated 360M attention.
- Per-layer gating matters ~3x more at 32 layers.** Per-layer NG-evict gains +0.015 and the per-layer oracle +0.032 over the shared gate at B=0.20 (3-layer Rung 1.5: +0.004 / +0.013; all $p=0.002$). Layer heterogeneity is real, headroom concentrates in the oracle, and the policy ordering is unchanged in all four views at every budget — deploy per-layer, evaluate either way.

Stated plainly (limitations)

One 360M base model on one corpus is still a single point in attention-distribution space; instruction-tuned and MoE-class models may differ, and tinyshakespeare prompts under-exercise long-range retrieval. Quality remains an attention-mass proxy, not end-task accuracy. $n=10$ keeps every Wilcoxon p at the two-sided floor (0.002): read as 'all ten traces agree in sign'. Capture ran on a constrained CPU sandbox (fp32 eager, validation path, not a speed path); traces are exact but few. Next rungs: end-task quality behind gates, larger trace counts with Diebold-Mariano testing, then Tier-2 production traces in the target serving stack.